

CLEARDALE AGDM, AB, Canada

WMO: 712210

Lat: 56.3144N

Lon: 119.7450W

Elev: 630

StdP: 93.98

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-33.9	-30.2	-37.7	0.1	-33.8	-33.7	0.2	-29.9	10.4	2.2	9.3	2.3	0.9	280	0.667

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	27.4	16.1	25.6	15.3	23.9	14.7	17.8	24.4	16.7	23.3	15.7	22.0	3.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	11.7	20.1	14.2	10.9	19.0	13.1	10.2	17.8	52.6	24.5	49.1	23.4	46.1	22.1	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.6	6.5	-38.0	30.6	3.1	2.0	-40.2	32.1	-42.0	33.2	-43.7	34.3	-45.9	35.8		
(5)			-38.0	19.6	3.0	1.6	-40.1	20.7	-41.9	21.7	-43.5	22.6	-45.7	23.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.7	-12.3	-11.2	-5.4	3.2	10.0	13.9	15.9	14.3	9.6	2.8	-7.7	-13.2		
	DBStd	12.27	9.18	8.18	7.54	4.78	4.18	2.95	2.58	3.25	4.14	4.75	7.29	8.37		
	HDD10.0	3563	692	594	477	208	52	3	0	4	57	224	532	719		
	HDD18.3	6076	951	828	736	455	260	136	82	128	262	480	782	977		
	CDD10.0	544	0	0	0	3	51	121	184	139	44	3	0	0		
	CDD18.3	15	0	0	0	0	0	3	8	4	0	0	0	0		
	CDH23.3	489	0	0	0	1	29	87	207	134	31	1	0	0		
	CDH26.7	88	0	0	0	0	3	12	41	28	4	0	0	0		
(14) Wind	WSAvg	2.5	2.2	2.0	2.4	2.9	2.9	2.7	2.7	2.4	2.7	2.6	2.1	2.0		
(15) Precipitation	PrecAvg	459	30	22	26	21	34	73	79	51	33	28	30	31		
	PrecMax	655	67	45	69	46	86	153	146	113	83	94	88	129		
	PrecMin	288	10	1	6	4	8	26	28	8	8	7	6	4		
	PrecStd	98	15	13	14	10	20	38	29	28	18	19	18	25		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	6.8	11.2	20.8	26.7	28.4	30.2	29.5	26.6	20.1	10.3	6.6		
		MCWB	3.4	3.2	5.1	9.8	13.5	16.1	17.8	16.7	14.6	11.4	5.5	2.9		
	2%	DB	5.6	5.2	8.5	16.5	23.4	25.7	27.6	26.7	23.4	15.0	7.0	4.6		
		MCWB	2.7	1.8	3.6	7.9	12.0	14.7	16.8	16.5	14.1	8.4	3.3	1.6		
	5%	DB	4.0	3.6	6.8	13.7	21.2	23.8	25.6	24.5	20.3	12.3	4.7	2.1		
		MCWB	1.5	0.8	2.5	6.2	10.7	14.1	16.0	15.2	12.5	7.0	1.7	-0.5		
	10%	DB	1.3	1.2	4.9	11.2	19.0	21.9	23.7	22.5	17.6	10.6	2.3	-0.3		
		MCWB	-0.7	-1.2	1.3	4.9	9.9	13.4	15.5	14.7	11.1	5.9	0.1	-2.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	3.6	5.7	10.4	14.5	17.9	19.9	18.7	16.5	12.1	5.7	3.3		
		MCDB	6.6	6.3	10.3	19.8	23.5	25.9	26.8	24.9	23.8	19.6	9.9	6.3		
	2%	WB	2.8	2.2	3.9	8.3	12.8	16.2	18.1	17.5	14.6	9.2	3.7	1.6		
		MCDB	5.5	4.8	8.1	16.1	21.7	23.0	24.5	24.1	21.8	13.8	6.7	4.5		
	5%	WB	1.5	0.8	2.7	6.5	11.6	15.1	17.1	16.3	13.1	7.5	1.8	-0.3		
		MCDB	3.8	3.4	6.4	12.6	19.4	21.7	23.4	22.9	19.4	11.5	4.3	1.7		
	10%	WB	-0.5	-1.2	1.4	5.2	10.5	14.0	16.1	15.2	11.7	6.2	0.2	-2.2		
		MCDB	1.5	1.2	4.7	10.5	17.7	20.2	22.2	21.1	16.8	10.0	1.9	-0.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	11.2	11.8	11.6	13.4	13.1	13.2	13.5	12.7	10.0	9.3	9.4		
		MCDBR	10.9	12.6	12.5	16.2	17.5	16.8	17.5	17.5	17.6	13.8	11.3	12.1		
		MCWBR	8.6	9.5	8.5	8.8	8.3	7.9	8.0	8.2	9.0	8.3	8.5	9.8		
	5% WB	MCDBR	10.8	12.0	11.9	15.2	15.9	14.3	15.5	15.5	17.0	13.3	10.9	11.9		
		MCWBR	8.6	9.1	8.2	8.5	7.8	7.1	8.0	8.1	9.0	8.4	8.5	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.254	0.287	0.320	0.346	0.348	0.350	0.346	0.306	0.284	0.240	0.219		
	taud		2.209	2.324	2.286	2.376	2.398	2.409	2.430	2.436	2.555	2.565	2.405	2.237		
	Ebn at Noon		659	834	882	901	893	895	884	865	862	800	703	623		
	Edh at Noon		53	71	96	103	108	108	104	97	74	57	43	39		
(44) All-Sky Solar Radiation	RadAvg		0.68	1.63	3.12	4.70	5.55	5.99	5.82	4.73	3.24	1.70	0.76	0.47		
	RadStd		0.07	0.13	0.23	0.24	0.38	0.51	0.34	0.42	0.27	0.20	0.07	0.03		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page